1.05 Linear Modelling

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What's the point of straight line equations?

We saw in Chapter 2 that lots of things in real life have a 'quadratic' relationship, e.g. vertical height with time. Lots of real life variables have a 'linear' relationship, i.e. **there is a fixed increase/decrease in one variable each time the other variable goes up by 1 unit.**

Examples

Car sales made and take home pay.



Temperature and altitude (in a particular location)



The relationship between Celsius and Fahrenheit.

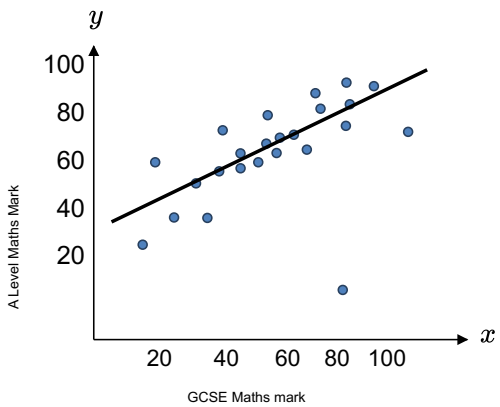


(And a pure maths one:)
The nth term of an arithmetic series.

3, 5, 8, 11, 14, …

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Modelling



A mathematical model is an attempt to model a real-life situation based on mathematical concepts.

For this example, **our model might be a linear model with equation $\mathbf{y=mx+c}$** where x is a student's GCSE mark and y is the predicted A Level mark.

Such a linear model can be drawn as a line of best fit.
The data obviously doesn't fit this line exactly. This chosen model may only partially fit the data (and the further the points are away from the line, the less suitable this model is).
We might decide another model, e.g. $y = ax^2 + bx + c$, is more appropriate.

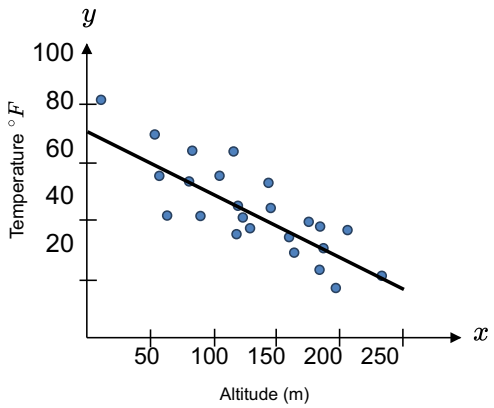
But if we choose a well-known model such as a linear one, then we can use **established mathematical theory** in useful ways:

- We need to choose the most appropriate 'parameters' m and c so the model best matches the data. You will learn in S1 there are existing techniques to do this.
- We can then predict a student's A Level mark based on their maths mark.

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Example



The temperature y at different points on a mountain is recorded at different altitudes x.
Suppose we were to use a linear model $y = mx + c$.

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a Determine m and c (you can assume the line goes through $(0, 70)$ and $(250, 20)$.

Interpret the meaning of m

b **The temperature at sea l** **n $0.2°\mathbf{F}$ for each extra**

c

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Evaluating if a linear model is most suitable

Choosing a model for our data usually involves making **simplifying assumptions**. For a linear model we are assuming:

The y value goes up the same amount for each unit increase in x.

The current population of Bickerstonia is 26000. This year (2017) the population increased by 150. Matt decides to model the population P based on the years t after 2017 by the linear model:

$$P = mt + c$$

Why might this not be a suitable model?

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Exercise 5H – Pure Y1 Page 106-108

3 The cost of electricity, E, in pounds and the number of kilowatt hours, h, are shown in the table.

kilowatt hours, h	0	15	40	60	80	110
cost of electricity, E	45	46.8	49.8	52.2	54.6	58.2

a Draw a graph of the data.
b Explain how you know a linear model would be appropriate.
c Deduce an equation in the form $E = ah + b$.
d Interpret the meaning of the coefficients a and b.
e Use the model to find the cost of 65 kilowatt hours.

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Exercise 5H – Pure Y1 Page 106-108

6 The average August temperature in Exeter is 20°C or 68°F. The average January temperature in the same place is 9°C or 48.2°F.
a Write an equation linking Fahrenheit F and Celsius C in the form $F = aC + b$. \hfill(3 marks)
b Interpret the values of a and b. \hfill(2 marks)
c The highest temperature recorded in the UK was 101.3°F. Calculate this temperature in Celsius.

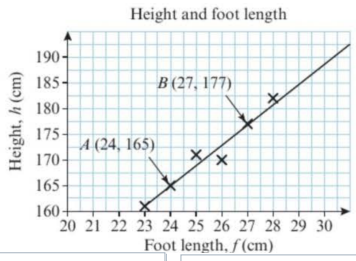
d For what value is the temperature in Fahrenheit the same as the temperature in Celsius?

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Exercise 5H – Pure Y1 Page 106-108

8 The scatter graph shows the height h and foot length f of 8 students. A line of best fit is drawn on the scatter graph.
a Explain why the data can be approximated to a linear model. \hfill(1 mark)
b Use points A and B on the scatter graph to write a linear equation in the form $h = af + b$. \hfill(3 marks)
c Calculate the expected height of a person with a foot le
26.5 cm. \hfill(1 mark)



9 The price P of a good and the quantity Q of a good are linked.
The demand for a new pair of trainers can be modelled using the equation $P = -\frac{3}{4}Q + 35$.
The supply of the trainers can be modelled using the equation $P = \frac{2}{3}Q + 1$.
a Draw a sketch showing the demand and supply lines on the same pair of axes.
The equilibrium point is the point where the supply and demand lines meet.
b Find the values of P and Q at the equilibrium point.